

Highlights

Disentangling the Effects of Stablecoins on Exchange Rates: A Currency-by-Currency Analysis

区分稳定币对汇率的影响：逐币种分析

Author Name

- Currency-by-currency daily panel for 27 economies, 2020–2025, with four stablecoin metrics per coin.
- 27 个经济体 2020–2025 年逐币种日度面板，每种稳定币含四类指标。
- Average effects of stablecoin growth on daily FX changes are small and mostly insignificant.
- 稳定币增长对日度汇率变化的平均效应很小且大多不显著。
- USDT coefficients are significant in developed and high-crypto-adoption economies, not in emerging markets.
- USDT 系数在发达经济体和高加密采用国家显著，在新兴市场不显著。
- Crypto-shock IV: strong first stage ($F = 74\text{--}248$) but insignificant second stage.
- 加密冲击工具变量：第一阶段强（ $F = 74\text{--}248$ ），第二阶段不显著。
- Circulation and trading volume dominate issuance and active wallets as transmission channels.
- 流通量与交易量是主导传导渠道，发行量与活跃钱包次之。

Disentangling the Effects of Stablecoins on Exchange Rates: A Currency-by-Currency Analysis

区分稳定币对汇率的影响：逐币种分析

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Abstract

We examine whether USD-pegged stablecoins (USDT, USDC, DAI) affect exchange rates across 27 economies at daily frequency from January 2020 to December 2025. Combining on-chain stablecoin metrics (circulation, issuance, transfer volume, and active addresses) with a multi-source panel of daily exchange rates and annual country characteristics, we estimate country-by-country panel regressions, standardized country-characteristic interactions, and a crypto-shock instrumental-variables design that exploits the Terra/LUNA collapse, the SVB-induced USDC de-peg, and lagged Ethereum network congestion. Average effects are small: conditional on global risk sentiment and broad dollar movements, stablecoin circulation growth has no significant average association with daily exchange-rate changes for USDT or DAI, while USDC circulation growth is associated with a modest local-currency appreciation of about -0.006 percentage points per 1% circulation growth. The heterogeneity runs opposite to the prevailing narrative: USDT coefficients are statistically significant in developed economies (0.0057) and in high-crypto-adoption countries (0.0098), but not in emerging markets. Instrumented estimates are insignificant for all three coins despite strong first stages (F between 74 and 248). At the daily horizon, stablecoin activity therefore appears to transmit to foreign exchange markets only weakly once endogeneity is addressed; the results caution against extrapolating crisis-episode evidence to normal times.

本文考察美元稳定币（USDT、USDC、DAI）是否影响 2020 年 1 月至 2025 年 12 月期间 27 个经济体的日度汇率。我们将链上稳定币指标（流通量、发行量、转账交易量、活跃地址数）与多来源的日度汇率和年度国家特征面板相结合，估计逐国面板回归、标准化国家特征交互项回归，并利用 Terra/LUNA 崩盘、硅谷银行引发的 USDC 脱锚以及滞后以太坊网络拥堵构建加密冲击工具变量设计。平均效应很小：在控制全球风险偏好与美元整体走势后，USDT 与 DAI 的流通量增长与日度汇率变化没有显著的平均关联，而 USDC 流通量每增长 1% 与本币约 0.006 个百分点的小幅升值相关。异质性方向与主流叙事相反：USDT 系数在发达经济体（0.0057）和高加密采用国家（0.0098）统计显著，而在新兴市场并不显著。尽管第一阶段很强（ F 介于 74 至 248），三种稳定币的工具变量估计均不显著。因此在日度频率上，一旦处理内生性，稳定币活动对外汇市场的传导似乎很弱；研究结果提示不应将危机时期的证据外推至正常时期。

Keywords: Stablecoins, Exchange rates, Panel data, Granular instrumental variables, Capital flows, Crypto adoption, *JEL classification:* F31, F38, G15, O16

1. Introduction / 引言

The rapid growth of USD-pegged stablecoins has raised important questions about their implications for the international monetary system and for foreign exchange (FX) markets in particular. Stablecoins were initially designed to facilitate crypto-asset trading, but they have increasingly become a parallel dollar market, offering users in countries with capital controls or currency instability an alternative avenue to access USD exposure [1, 2]. A growing literature documents that stablecoin demand interacts with safe-asset prices [3], with the stability of the peg itself [4], and with cross-border capital flows [5]. Whether stablecoin activity moves national exchange rates in normal times, and for which currencies, remains largely open.

与美元挂钩的稳定币的快速增长，引发了关于其对国际货币体系、尤其是对外汇市场 (FX) 影响的重要问题。稳定币最初旨在便利加密资产交易，但它们已日益成为一个平行的美元市场，为资本管制或币值不稳国家的用户提供了获取美元敞口的替代渠道 [1, 2]。越来越多的文献表明，稳定币需求会与资产价格 [3]、锚定稳定性本身 [4] 以及跨境资本流动 [5] 相互作用。然而，在正常时期稳定币活动是否会推动各国汇率变动、作用于哪些货币，在很大程度上仍是悬而未决的问题。

This paper provides a comprehensive currency-by-currency analysis of how stablecoins relate to exchange rates. We make three contributions. First, we disentangle the effects of individual stablecoins—USDT, USDC, and DAI—rather than treating stablecoins as an aggregate. This distinction matters because the three coins differ in collateral structure, user base, and regulatory status, which likely leads to heterogeneous links with FX markets. Second, we construct four complementary indicators for each coin—circulation (stock), issuance (flow), transfer volume (activity), and active addresses (adoption)—which allows us to identify the dominant transmission channel. Third, we address endogeneity with a crypto-shock instrumental-variable (IV) strategy in the spirit of granular instrumentation [6, 7], exploiting the Terra/LUNA collapse of May 2022, the USDC de-peg around the Silicon Valley Bank failure of March 2023, and lagged Ethereum gas-price changes as sources of variation in stablecoin circulation that are plausibly orthogonal to country-specific FX fundamentals.

本文对稳定币与汇率的关系进行了全面的逐币种分析，贡献有三。第一，我们区分单个稳定币——USDT、USDC 与 DAI——而非将稳定币作为一个整体处理。这一区分很重要，因为三种币在抵押结构、用户基础和监管地位上存在差异，很可能导致其与外汇市场的关联各不相同。第二，我们为每种币构建了四个互补指标——流通量（存量）、发行量（流量）、转账交易量（活跃度）与活跃地址数（采用度）——从而能够识别主导的传导渠道。第三，我们借鉴工具变量的“颗粒化”思想 [6, 7]，采用加密冲击工具变量策略处理内生性，利用 2022 年 5 月 Terra/LUNA 崩盘、2023 年 3 月硅谷银行倒闭前后的 USDC 脱锚以及滞后以太坊 Gas 价格变动，作为稳定币流通量变异的外生来源，这些来源与各国汇率基本面可合理地认为正交。

Our analysis covers 27 economies—11 advanced and 16 emerging—at daily frequency from January 2020 to December 2025. Because stablecoin metrics vary only over time, they are absorbed by time fixed effects; our baseline therefore conditions on global controls (VIX changes and broad dollar index changes) and relies on country fixed effects, while interactions with standardized country characteristics restore cross-sectional identification along the dimensions suggested by the literature: capital account openness [8], exchange-rate regime [9], external buffers, fiscal space, income level, and crypto adoption [10].

我们的分析覆盖 27 个经济体——11 个发达经济体和 16 个新兴经济体——频率为日度，样本期为 2020 年 1 月至 2025 年 12 月。由于稳定币指标仅随时间变化，会被时间固定效应完全吸收；因此基准模型以全球控制变量（VIX 变动与广义美元指数变动）为条件，依赖国家固定

效应，而与标准化国家特征的交互项则沿文献提示的维度恢复截面识别：资本账户开放度 [8]、汇率制度 [9]、外部缓冲、财政空间、收入水平与加密采用度 [10]。

The headline findings are sobering for the view that stablecoins are a first-order driver of exchange rates in normal times. At the daily horizon, average effects are small and mostly insignificant. Where effects appear, they run opposite to the prevailing narrative: USDT circulation growth is associated with local-currency depreciation in developed economies and in high-crypto-adoption countries, but not in emerging markets; USDC circulation growth is associated with modest local-currency appreciation in the full sample and in several subsamples. The IV design confirms that crypto-native shocks strongly move stablecoin circulation (first-stage F -statistics of 74 to 248) but that this exogenous variation has no detectable average effect on daily exchange-rate changes. Crisis-episode spillovers documented elsewhere thus do not appear to generalize to normal times at the daily frequency.

总体结论对“稳定币在正常时期是汇率一阶驱动因素”的观点而言并不乐观。在日度频率上，平均效应很小且大多不显著。在效应出现之处，其方向与主流叙事相反：USDT 流通量增长与发达经济体和高加密采用国家的本币贬值相关，而在新兴市场并不相关；USDC 流通量增长则与全样本及若干子样本中本币的小幅升值相关。工具变量设计证实，加密原生冲击确实强烈影响稳定币流通量（第一阶段 F 统计量为 74 至 248），但这种外生变异对日度汇率变化没有可检测的平均效应。因此，其他文献记载的危机时期溢出效应似乎并不能推广到正常时期的日度频率。

The rest of the paper is organized as follows. Section 2 provides institutional background and a conceptual framework. Section 3 describes the data and variable construction. Section 4 outlines the empirical strategy. Section 5 presents the main results, Section 6 the heterogeneity analysis, and Section 7 the robustness checks. Section 8 concludes.

本文其余部分安排如下：第 2 节介绍制度背景与概念框架；第 3 节描述数据与变量构造；第 4 节阐述实证策略；第 5 节报告主要结果；第 6 节进行异质性分析；第 7 节进行稳健性检验；第 8 节总结全文。

2. Institutional Background and Conceptual Framework / 制度背景与概念框架

2.1. The Stablecoin Ecosystem / 稳定币生态

USD-pegged stablecoins are digital tokens designed to maintain a one-to-one exchange rate with the US dollar. Although they share the objective of dollar stability, the three largest coins differ markedly in structure and user base [1]. USDT (Tether), launched in 2014, is the oldest and largest stablecoin; it dominates retail crypto trading and is frequently used as a dollar substitute in emerging markets. USDC (USD Coin), launched in 2018 and issued by Circle, is backed by cash and short-dated US Treasury securities and is favored by institutional users in open, developed economies. DAI, launched in 2019 and maintained by the MakerDAO (Sky) protocol, is backed by crypto collateral and is used primarily within the DeFi ecosystem, so its dynamics reflect crypto-market sentiment more than cross-border capital flows [4, 11].

与美元挂钩的稳定币是旨在与美元维持一比一汇率的数字代币。尽管目标一致，三大稳定币在结构和用户基础上差异显著 [1]。USDT (Tether) 于 2014 年推出，是历史最久、规模最大的稳定币，主导零售加密交易，并在新兴市场常被用作美元替代品。USDC (USD Coin) 于 2018 年由 Circle 发行，以现金和短期美国国债为储备，受到开放发达经济体中机构用户的青睐。DAI 于 2019 年推出，由 MakerDAO (Sky) 协议维护，以加密资产为抵押，主要用于 DeFi 生态，其动态更多反映加密市场情绪而非跨境资本流动 [4, 11]。

2.2. Conceptual Framework / 概念框架

We identify four potential channels through which stablecoins may affect exchange rates. First, the *asset substitution channel*: when residents increase their holdings of dollar stablecoins (circulation), they substitute away from domestic-currency assets, and the additional dollar exposure may put depreciation pressure on the domestic currency. Second, the *supply shock channel*: net issuance creates new dollar-denominated digital assets, and large issuance may expand dollar liquidity in the crypto ecosystem with spillovers to traditional FX markets. Third, the *trading flow channel*: high transfer volume on stablecoin–fiat pairs indicates intensified currency conversion that may affect FX order flow. Fourth, the *network adoption channel*: growth in active addresses signals expanding adoption that facilitates capital movements outside traditional payment systems [5].

我们识别了稳定币可能影响汇率的四条潜在渠道。第一是资产替代渠道：当居民增持美元稳定币（流通量）时，他们替代了本币资产，新增的美元敞口可能对本币形成贬值压力。第二是供给冲击渠道：净发行创造新的美元计价数字资产，大规模发行可能扩张加密生态中的美元流动性并溢出至传统外汇市场。第三是交易流渠道：稳定币—法币交易对的高转账交易量表明货币兑换活动加剧，可能影响外汇市场订单流。第四是网络采用渠道：活跃地址数的增长标志着采用度扩大，从而促进游离于传统支付体系之外的资本流动 [5]。

These channels imply testable heterogeneity. If asset substitution dominates, effects should be strongest where demand for dollar substitutes is highest—plausibly emerging markets with capital controls or weak external positions (our hypothesis H1 and H5). If institutional usage dominates, USDC should matter more in developed, financially open economies (H2). If crypto sentiment dominates, DAI effects should be weak and confined to high-adoption countries (H3). Across metrics, circulation and volume should dominate issuance and wallets (H4), and fiscally constrained countries should be more sensitive (H6).

这些渠道蕴含可检验的异质性。若资产替代渠道主导，效应应在美元替代品需求最强之处——很可能是实施资本管制或外部头寸较弱的新兴市场——最为显著（假说 H1 与 H5）。若机构使用主导，USDC 应在发达且金融开放的经济体中影响更大（H2）。若加密情绪主导，DAI 的效应应较弱且仅限于高采用国家（H3）。就指标而言，流通量与交易量应主导发行量与钱包数（H4）；财政受限国家对稳定币冲击应更敏感（H6）。

3. Data and Variable Construction / 数据与变量构造

3.1. Data Sources / 数据来源

Our analysis combines plugin-fetched data from multiple providers, all archived in the replication package (`Countries Dataset/`, located at the project root alongside `Manuscript1/`).

我们的分析整合了通过数据插件从多个供应商获取的数据，全部存档于复现数据包 (`Countries Dataset/`，位于项目根目录，与 `Manuscript1/` 同级)。

Stablecoin data / 稳定币数据. Daily on-chain metrics for USDT, USDC, and DAI come from Glassnode (accessed July 2026): circulating supply, aggregated transfer volume, and aggregated active addresses [12].

USDT、USDC 与 DAI 的日度链上指标来自 Glassnode（2026 年 7 月获取）：流通供给量、汇总转账交易量与汇总活跃地址数 [12]。

Exchange rates / 汇率. Daily spot rates against the USD are assembled from three complementary sources to maximize coverage. Thirteen currencies come directly from FRED (JPY, CAD, CHF, SEK, NOK, MXN, BRL, INR, CNY, KRW, SGD, MYR, ZAR); EUR and GBP are inverted from FRED's USD-per-currency quotes. Seven currencies (AUD, IDR, PHP, PLN, THB, TRY, and RUB until February 2022) are converted from ECB euro reference rates, and five (ARS, CLP, COP, NGN, SAR, plus RUB from March 2022 onward) come from Yahoo Finance. All rates are expressed as local currency units per USD, so an increase denotes a depreciation of the local currency. The sample spans 2 January 2020 to 31 December 2025 and is an unbalanced panel of 40,667 currency-days.

美元兑各货币的日度即期汇率由三个互补来源拼接而成，以最大化覆盖面。十三种货币直接取自 FRED (JPY、CAD、CHF、SEK、NOK、MXN、BRL、INR、CNY、KRW、SGD、MYR、ZAR)；EUR 与 GBP 由 FRED 的美元兑外币报价取倒数得到。七种货币 (AUD、IDR、PHP、PLN、THB、TRY，以及 2022 年 2 月之前的 RUB) 由欧洲央行欧元参考汇率换算，五种货币 (ARS、CLP、COP、NGN、SAR，以及 2022 年 3 月起的 RUB) 来自雅虎财经。所有汇率均以每美元兑本币单位数表示，数值上升表示本币贬值。样本覆盖 2020 年 1 月 2 日至 2025 年 12 月 31 日，为包含 40,667 个“货币一日”观测的非平衡面板。

Country characteristics / 国家特征. Annual moderators combine the World Bank WDI (GDP per capita, CPI inflation, reserves in months of imports, current account % GDP), the IMF WEO (general government gross debt % GDP, fiscal balance % GDP), and two academic datasets: the Chinn–Ito capital account openness index [8] and the Ilzetzi–Reinhart–Rogoff (IRR) coarse exchange-rate regime classification [9]. Euro-area aggregates for government debt and the fiscal balance come from Eurostat (EA20), and the euro-area current account from the IMF DataMapper euro-area aggregate. Crypto adoption is measured with the Chainalysis Global Crypto Adoption Index [10], coded as a score of $(21 - r)/20$ for a country ranked r in the annual top-20 and zero otherwise. Three gaps in the WDI required adjustments: private credit (% GDP) is discontinued for Canada, Switzerland, and Saudi Arabia, WDI external debt is not collected for high-income economies, and Nigeria's trade share is missing; the moderator set therefore uses fiscal balance and log GDP per capita, which are fully covered. Short within-country gaps are closed by carry-forward; the Chinn–Ito index (ending 2023) and the latest IRR classification are held constant thereafter. China-specific series (foreign reserves, CPI, and the CSI 300 index, used for extensions) were retrieved from Wind; iFind was used for cross-checks but its historical interface covers individual equities only.

年度调节变量整合了世界银行 WDI (人均 GDP、CPI 通胀、外汇储备覆盖进口月数、经常账户占 GDP 比重)、IMF WEO (政府总债务占 GDP、财政余额占 GDP)，以及两套学术数据集：Chinn–Ito 资本账户开放度指数 [8] 与 Ilzetzi–Reinhart–Rogoff (IRR) 汇率制度粗略分类 [9]。欧元区政府债务与财政余额取自 Eurostat (EA20 口径)，欧元区经常账户取自 IMF DataMapper 的欧元区汇总值。加密采用度用 Chainalysis 全球加密采用指数 [10] 度量：年度前 20 名中排名第 r 的国家赋分为 $(21 - r)/20$ ，未上榜国家记为零。WDI 数据存在三处缺口需作调整：加拿大、瑞士、沙特的私人信贷 (占 GDP 比重) 序列已停止更新；WDI 不收集高收入经济体的外债数据；尼日利亚的贸易占比缺失。因此调节变量集采用覆盖完整的财政余额与人均 GDP 对数。国内的短期缺口用前值结转填补；Chinn–Ito 指数 (止于 2023 年) 与最新的 IRR 分类在之后年份保持不变。中国相关序列 (外汇储备、CPI、沪深 300 指数，用于扩展分析) 取自 Wind；iFind 用于交叉核验，但其历史数据接口仅覆盖个股。

Global controls / 全球控制变量. Global controls are the daily change in the VIX and in the Federal Reserve's broad trade-weighted dollar index (both from FRED), with the DXY used in

robustness.

全球控制变量为 VIX 与美联储广义贸易加权美元指数的日度变动（均取自 FRED），稳健性检验中改用 DXY。

3.2. Variable Construction / 变量构造

The dependent variable is the daily log change of the exchange rate:

$$\Delta s_{i,t} = \ln(S_{i,t}) - \ln(S_{i,t-1}), \quad (1)$$

where $S_{i,t}$ is the spot rate of country i on day t in local currency per USD; positive values denote depreciation. To limit the influence of data errors and extreme prints, $\Delta s_{i,t}$ is winsorized at the 1st and 99th percentiles within each currency.

被解释变量为汇率的日度对数变化：其中 $S_{i,t}$ 为第 i 国第 t 日的即期汇率（本币/美元），取正值表示本币贬值。为限制数据误差与极端值的影响， $\Delta s_{i,t}$ 在每种货币内部按 1% 与 99% 分位缩尾。

For each stablecoin $j \in \{\text{USDT}, \text{USDC}, \text{DAI}\}$ we construct four daily metrics: the log change in circulation $\Delta \ln(\text{Circ}_{j,t})$; a signed log change in net issuance, $\Delta[\text{sign}(I_{j,t}) \ln(1 + |I_{j,t}|)]$ with $I_{j,t}$ the net issuance in billions of USD, which accommodates negative issuance; the log change in transfer volume $\Delta \ln(\text{Vol}_{j,t})$; and the log change in active addresses $\Delta \ln(\text{Wallets}_{j,t})$.

对每种稳定币 $j \in \{\text{USDT}, \text{USDC}, \text{DAI}\}$ ，我们构造四个日度指标：流通量对数变化 $\Delta \ln(\text{Circ}_{j,t})$ ；净发行的带符号对数变化 $\Delta[\text{sign}(I_{j,t}) \ln(1 + |I_{j,t}|)]$ （其中 $I_{j,t}$ 为以十亿美元计的净发行，该变换可处理负发行）；转账交易量对数变化 $\Delta \ln(\text{Vol}_{j,t})$ ；以及活跃地址数对数变化 $\Delta \ln(\text{Wallets}_{j,t})$ 。

Country characteristics enter as lagged annual values (year $t - 1$ values apply to year t) and, for interaction regressions, as standardized 2023 cross-sectional values $z(\text{Char}_i)$.

国家特征以滞后一年的年度值进入模型（第 t 年使用第 $t - 1$ 年的值），在交互项回归中则使用 2023 年截面的标准化值 $z(\text{Char}_i)$ 。

Table 1 reports summary statistics. Daily exchange-rate changes average close to zero with substantially higher volatility in emerging markets (standard deviation 0.77%) than in developed economies (0.56%). Stablecoin metrics are volatile: mean daily circulation growth is about 0.2% for all three coins, while issuance, volume, and wallet growth are an order of magnitude noisier. The 2023 cross-section of country characteristics shows wide dispersion in capital openness, reserve coverage (1.3 to 16.6 months of imports), government debt (15% to 220% of GDP), and crypto adoption.

表 1 报告了描述性统计。日度汇率变化均值接近于零，新兴市场的波动率（标准差 0.77%）明显高于发达经济体（0.56%）。稳定币指标波动剧烈：三种币的日均流通量增速约为 0.2%，而发行量、交易量与钱包数增速的噪声要高出一个数量级。2023 年国家特征截面显示，资本开放度、储备覆盖率（1.3 至 16.6 个月进口）、政府债务（占 GDP 的 15% 至 220%）与加密采用度的离散程度都很大。

Figure 1 plots indexed exchange rates for selected currencies, Figure 2 shows stablecoin supply and 30-day circulation growth against the dollar index, and Figure 3 summarizes the dispersion of the four key country characteristics.

图 1 绘制了部分货币的汇率指数走势，图 2 展示了稳定币供给量与 30 日流通量增速同美元指数关系，图 3 概括了四项关键国家特征的分布离散程度。

Exchange rates against the USD, 2020-2025

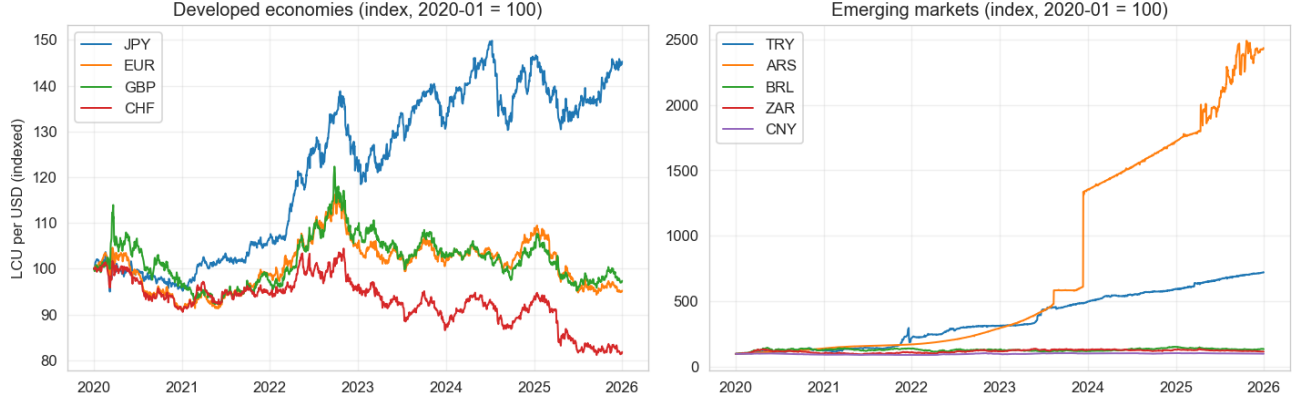


Figure 1: Exchange rates against the USD, 2020–2025 (index, January 2020 = 100) / 兑美元汇率走势, 2020–2025 (指数化, 2020 年 1 月 = 100) . Left: developed economies; right: emerging markets.

Stablecoin metrics and the USD index

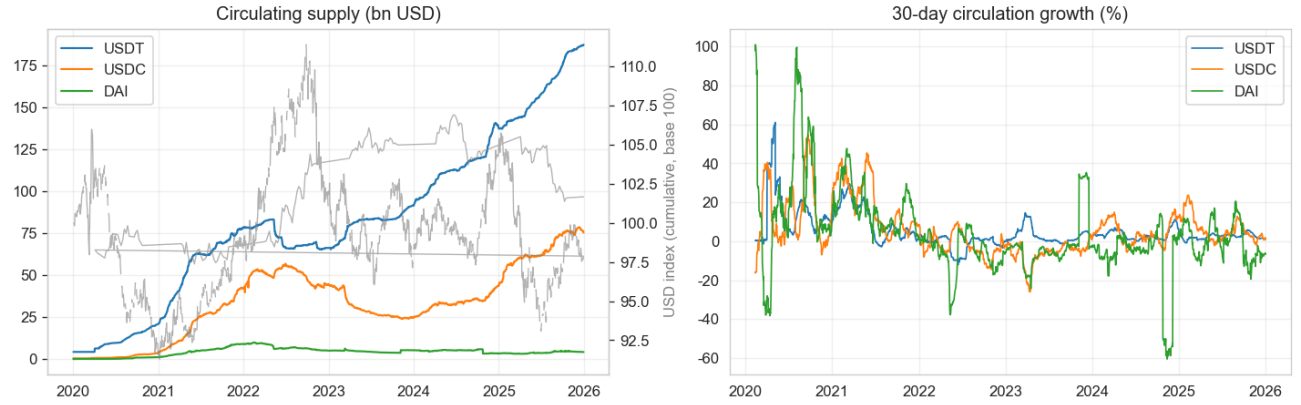


Figure 2: Stablecoin supply and 30-day circulation growth / 稳定币供给量与 30 日流通量增速. Left axis: circulating supply (bn USD) with the USD index (grey, right axis); right panel: 30-day circulation growth (%).

4. Empirical Methodology / 实证方法

4.1. Baseline Specification / 基准设定

We estimate panel regressions of daily exchange-rate changes on stablecoin metrics:

$$\Delta s_{i,t} = \alpha_i + \sum_{j \in \{\text{USDT, USDC, DAI}\}} \sum_{k \in \{\text{Circ, Iss, Vol, Wallets}\}} \beta_{j,k} \Delta \ln(X_{j,t}^k) + \gamma' Z_t + \epsilon_{i,t}, \quad (2)$$

where α_i are country fixed effects, $X_{j,t}^k$ is metric k of coin j , and Z_t contains the daily VIX change and broad dollar index change. Standard errors are clustered by country.

我们估计日度汇率变化对稳定币指标的面板回归：其中 α_i 为国家固定效应， $X_{j,t}^k$ 为稳定币 j 的第 k 项指标， Z_t 包含 VIX 日变动与广义美元指数日变动。标准误在国家层面聚类。

Because stablecoin metrics vary only over time, they are perfectly collinear with time fixed effects; equation (2) therefore replaces time effects with global controls and identifies $\beta_{j,k}$ from

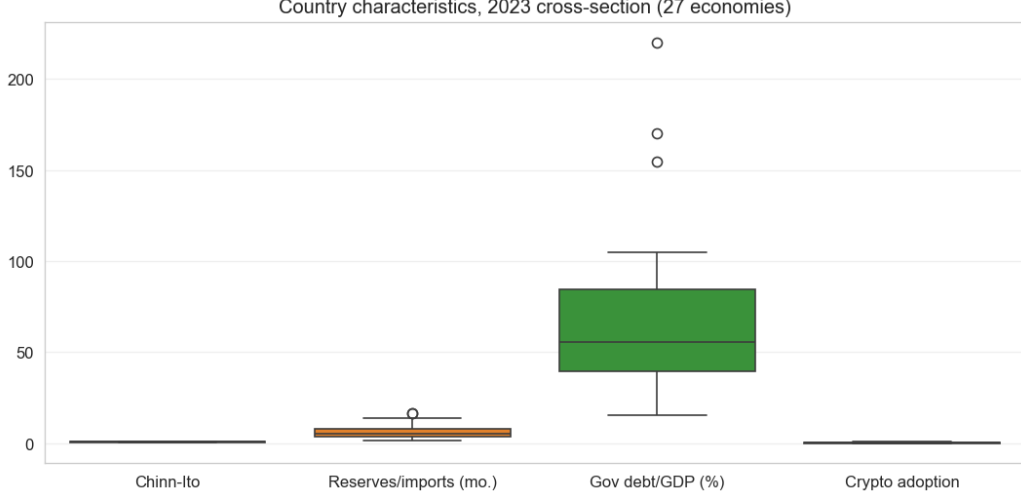


Figure 3: Country characteristics, 2023 cross-section (27 economies) / 国家特征分布，2023 年截面（27 个经济体）。

time-series comovement, conditional on global risk sentiment and broad dollar strength. Cross-sectional identification is restored by interacting the metrics with country characteristics below. All specifications are estimated with `linearmodels` and cross-checked with the `statspai` package.

由于稳定币指标仅随时间变化，它们与时间固定效应完全共线；因此式 (2) 以全球控制变量替代时间效应，并在控制全球风险偏好与美元整体走势的条件下，通过时间序列联动识别 $\beta_{j,k}$ 。下文通过稳定币指标与国家特征的交互恢复截面识别。所有设定均用 `linearmodels` 估计，并用 `statspai` 包交叉核验。

4.2. Country-Characteristic Interactions / 国家特征交互项

To test whether the sensitivity of exchange rates to stablecoin activity varies systematically with country characteristics, we estimate

$$\Delta s_{i,t} = \alpha_i + \beta_1 \Delta \ln(\text{Circ}_{\text{USDT},t}) + \beta_2 [\Delta \ln(\text{Circ}_{\text{USDT},t}) \times z(\text{Char}_i)] + \delta' \Delta \ln(\text{Circ}_{-j,t}) + \gamma' Z_t + \epsilon_{i,t}, \quad (3)$$

where $z(\text{Char}_i)$ is the standardized 2023 value of one of eight characteristics: capital openness, a fixed-regime indicator, reserve coverage, government debt, the fiscal balance, the current account, income level, and crypto adoption. A positive β_2 means that countries with higher values of the characteristic show a more pronounced depreciation response to USDT growth.

为检验汇率对稳定币活动的敏感度是否随国家特征系统性变化，我们估计：其中 $z(\text{Char}_i)$ 为八项特征之一的 2023 年标准化取值：资本开放度、固定汇率制度哑变量、储备覆盖、政府债务、财政余额、经常账户、收入水平与加密采用度。 β_2 为正意味着该特征取值越高的国家，其汇率对 USDT 增长的贬值反应越强。

4.3. Crypto-Shock Instrumental Variables / 加密冲击工具变量

Reverse causality is a first-order concern: anticipated depreciation may itself raise demand for dollar stablecoins. We therefore instrument circulation growth with three crypto-native shocks that are plausibly orthogonal to country-specific FX fundamentals: (i) a dummy for the Terra/LUNA collapse window (5–20 May 2022); (ii) a dummy for the USDC de-peg window

around the Silicon Valley Bank failure (9–17 March 2023); and (iii) the lagged daily change in the mean Ethereum gas price, which captures network congestion shocks [11]. The design follows the granular-IV logic of using idiosyncratic, economy-level shocks to generate exogenous variation [6, 7]. The first stage is

$$\Delta \ln(\text{Circ}_{j,t}) = \pi_0 + \pi_1 \text{Terra}_t + \pi_2 \text{SVB}_t + \pi_3 \Delta \ln(\text{Gas}_{t-1}) + \Pi' W_t + \nu_{j,t}, \quad (4)$$

and the second stage replaces circulation growth in a by-coin version of (2) with its fitted value. Country fixed effects are removed by within transformation and standard errors are clustered by country. We report partial first-stage F -statistics from a joint test of the three excluded instruments.

反向因果是一阶关切：对贬值的预期本身可能推高对美元稳定币的需求。因此我们用三个加密原生冲击作为流通量增长的工具变量，它们与各国汇率基本面可合理地认为正交：(i) Terra/LUNA 崩盘窗口哑变量（2022 年 5 月 5 日至 20 日）；(ii) 硅谷银行倒闭前后 USDC 脱锚窗口哑变量（2023 年 3 月 9 日至 17 日）；(iii) 以太坊平均 Gas 价格日变动的滞后值，用以捕捉网络拥堵冲击 [11]。该设计遵循颗粒化工具变量利用特质性冲击生成外生变异的逻辑 [6, 7]。第一阶段为第二阶段则在分币种版式 (2) 中以流通量增长的拟合值替代原值。国家固定效应通过组内变换消除，标准误在国家层面聚类。我们报告对三个排他性工具变量联合检验的第一阶段偏 F 统计量。

5. Main Results / 主要结果

5.1. Baseline Results / 基准结果

Table 2 presents the baseline estimates. Column (1) pools the three coins into a supply-weighted aggregate circulation measure; column (2) disaggregates by coin; column (3) includes all four metrics per coin.

表 2 报告基准估计。第 (1) 列将三种币合成为按供给加权的总流通量指标；第 (2) 列按币种分解；第 (3) 列纳入每种币的全部四项指标。

Two facts stand out. First, the broad dollar index is by far the dominant driver of daily bilateral moves: a 1% rise in the broad dollar index is associated with an average bilateral depreciation of about 0.92%, and global controls together with country fixed effects explain about 20% of daily variation. Second, stablecoin activity matters little on average. The aggregate coefficient in column (1) is small and insignificant (0.0035, s.e. 0.0035). In column (2), none of the three coin-level circulation coefficients is significant. In the saturated column (3), USDC circulation growth enters negatively and significantly (-0.0063 , s.e. 0.0029): conditional on the other metrics, faster USDC growth coincides with modest local-currency *appreciation*. DAI transfer volume is also negative and significant (-0.0002 , s.e. 0.0001), while DAI issuance is marginally positive. All coefficients are economically tiny relative to the 0.69% standard deviation of daily FX changes.

有两点事实尤为突出。第一，广义美元指数是日度双边汇率变动的绝对主导因素：广义美元指数每上升 1%，双边汇率平均贬值约 0.92%；全球控制变量与国家固定效应合计解释约 20% 的日度波动。第二，稳定币活动的平均影响很小。第 (1) 列的汇总系数很小且不显著 (0.0035, 标准误 0.0035)；第 (2) 列三种币的流通量系数均不显著；在饱和的第 (3) 列中，USDC 流通量增速显著为负 (-0.0063 , 标准误 0.0029)，即在控制其他指标后，USDC 增长越快，本币反而小幅升值。DAI 交易量同样显著为负 (-0.0002 , 标准误 0.0001)，DAI 发行量边际显著为正。相对于日度汇率变化 0.69% 的标准差，所有系数在经济意义上都很小。

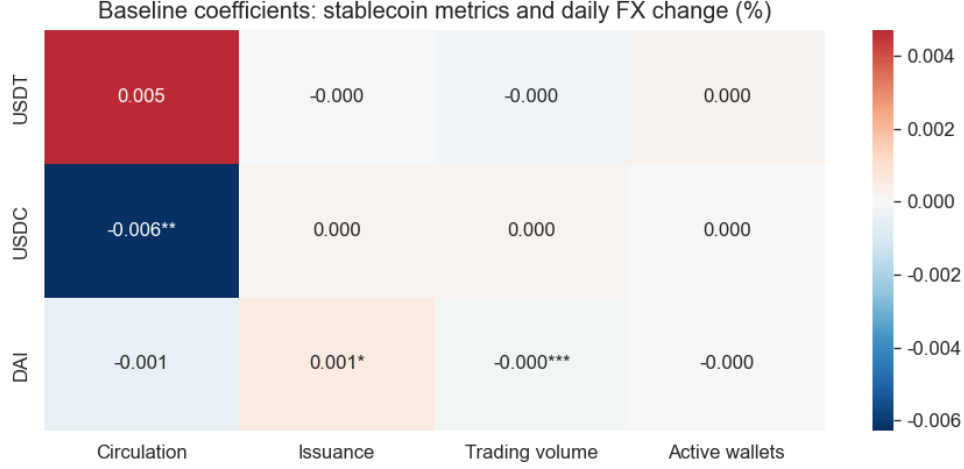


Figure 4: Baseline coefficients by coin and metric / 分币种分指标的基准系数. Annotations report point estimates with significance stars; red (blue) denotes positive (negative) coefficients.

Figure 4 visualizes the full set of coefficients from column (3). The pattern—a negative USDC circulation effect, a negative DAI volume effect, and insignificant USDT coefficients—suggests that at daily frequency, stablecoin flows co-move more with broad dollar strength and crypto-market conditions than with country-specific depreciation pressure.

图 4 直观展示了第 (3) 列的全部系数。负的 USDC 流通量效应、负的 DAI 交易量效应以及不显著的 USDT 系数共同表明：在日度频率上，稳定币流动更多与美元整体走势和加密市场状况同步波动，而非与针对特定国家的贬值压力相关。

5.2. Country-Characteristic Interactions / 国家特征交互项结果

Table 3 reports equation (3) for the eight standardized characteristics. Only two interactions are significant at the 10% level. The positive reserves interaction (0.0037, s.e. 0.0019) implies that countries with *stronger* reserve buffers show a somewhat more positive (depreciation) response of the exchange rate to USDT growth, which runs counter to the external-vulnerability hypothesis H5. The negative fiscal-balance interaction (-0.0024 , s.e. 0.0013) indicates that countries with larger fiscal surpluses exhibit a less positive response, consistent with limited sensitivity of fiscally strong countries but not with the strong form of H6. Interactions with capital openness, the exchange-rate regime, government debt, the current account, income, and crypto adoption are all insignificant. Overall, systematic moderation by country characteristics is weak at the daily frequency.

表 3 报告了八项标准化特征的式 (3) 估计结果。仅有两个交互项在 10% 水平显著。储备交互项为正 (0.0037, 标准误 0.0019)，意味着储备缓冲越强的国家，其汇率对 USDT 增长的贬值反应反而略强，这与外部脆弱性假说 H5 相反。财政余额交互项为负 (-0.0024 , 标准误 0.0013)，表明财政盈余越大的国家反应越弱，这与财政强国敏感度有限相一致，但不支持强形式的 H6。资本开放度、汇率制度、政府债务、经常账户、收入水平与加密采用度的交互项均不显著。总体而言，日度频率上国家特征对效应的系统性调节很弱。

5.3. Instrumental Variable Results / 工具变量结果

Table 4 presents the IV estimates. The first stage is strong: joint F -statistics for the three excluded instruments are 74.0 (USDT), 248.2 (USDC), and 75.3 (DAI), far above conventional thresholds, confirming that crypto-native shocks move stablecoin circulation. The second stage,

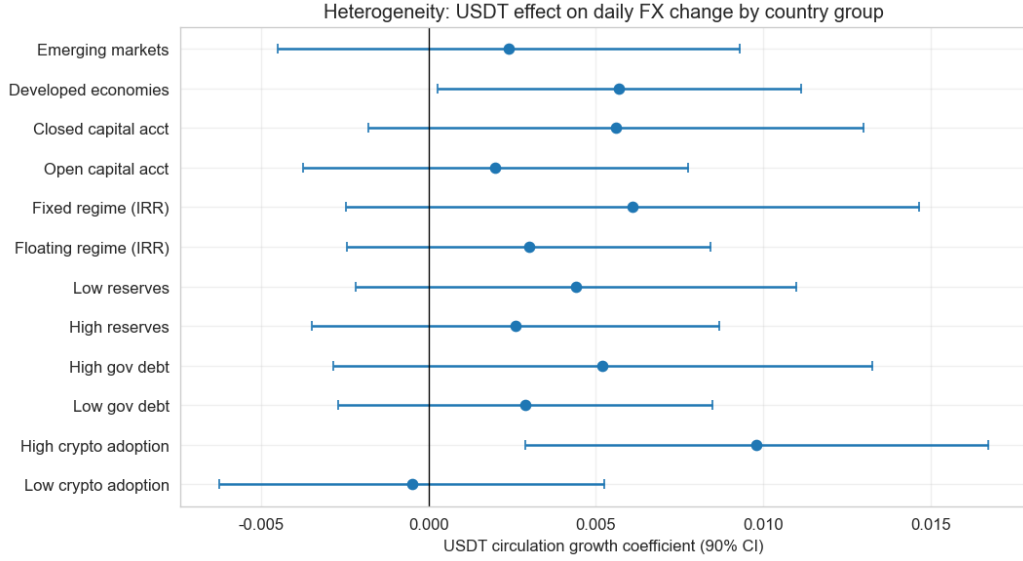


Figure 5: Heterogeneity in the USDT circulation coefficient / USDT 流通量系数的异质性, by country group (90% confidence intervals).

however, delivers insignificant coefficients for all three coins: -0.019 (s.e. 0.059) for USDT, 0.024 (s.e. 0.026) for USDC, and 0.021 (s.e. 0.018) for DAI. Under the exclusion restriction that these shocks affect bilateral exchange rates only through stablecoin circulation, there is no detectable causal effect of stablecoin circulation on daily FX changes at the sample average. The (insignificant) IV point estimates are larger in absolute value than their OLS counterparts, consistent with substantial attenuation from reverse causality and measurement error in the daily comovement, but the uncertainty is too large to draw magnitude conclusions.

表 4 报告工具变量估计。第一阶段很强：三个排他性工具变量的联合 F 统计量分别为 74.0 (USDT)、248.2 (USDC) 和 75.3 (DAI)，远高于常规门槛，证实加密原生冲击确实推动稳定币流通量。然而第二阶段三种币的系数均不显著：USDT 为 -0.019 (标准误 0.059)，USDC 为 0.024 (标准误 0.026)，DAI 为 0.021 (标准误 0.018)。在“这些冲击仅通过稳定币流通量影响双边汇率”的排他性约束下，样本平均意义上不存在可检测的稳定币流通量对日度汇率变化的因果效应。(不显著的) IV 点估计的绝对值大于 OLS 对应值，这与日度联动中反向因果与测量误差造成的衰减一致，但不确定性太大，无法就量级下结论。

6. Heterogeneity Analysis / 异质性分析

Table 5 splits the sample along six dimensions; Figure 5 plots the USDT coefficients with 90% confidence intervals.

表 5 沿六个维度对样本分组；图 5 绘制了 USDT 系数及其 90% 置信区间。

Three regularities emerge. First, against H1, the USDT coefficient is *not* significant in emerging markets; it is significant in developed economies (0.0057) and, most strongly, in high-crypto-adoption countries (0.0098). The asset-substitution narrative for emerging markets is thus not visible in daily data over 2020–2025; instead, USDT’s measurable footprint appears where crypto market infrastructure is deepest. Second, the USDC coefficient is negative in most groups and significant in developed economies, fixed-regime, and low-reserve countries—consistent with USDC growth coinciding with flight-to-quality dollar strength rather than depreciation of the

sampled currencies. Third, DAI effects are small and sign patterns vary across groups, in line with H3's weak-effect prediction.

可以归纳出三点规律。第一，与 H1 相反，USDT 系数在新兴市场并不显著，而在发达经济体（0.0057）、尤其在高加密采用国家（0.0098）显著。因此 2020–2025 年的日度数据中看不到新兴市场的资产替代叙事；USDT 可测量的足迹出现在加密市场基础设施最完善之处。第二，USDC 系数在多数分组中为负，并在发达经济体、固定汇率制和低储备国家显著——这与 USDC 增长伴随避险性美元走强、而非样本货币贬值相一致。第三，DAI 效应较小且符号随分组变化，符合 H3 的弱效应预测。

7. Robustness Checks / 稳健性检验

Table 6 summarizes four checks; full estimates are in the replication files.

表 6 汇总了四项检验；完整估计见复现文件。

First, at weekly frequency all coin coefficients are insignificant: the small daily comovements do not survive aggregation, indicating that any stablecoin–FX link is a short-horizon phenomenon. Second, excluding Russia and the Terra/SVB crisis months leaves the negative USDC circulation coefficient significant (-0.0051), showing that the baseline result is not driven by these episodes. Third, replacing the broad dollar index with the DXY flips the USDT coefficient to a marginally significant negative value (-0.0051), underscoring that coin coefficients are sensitive to how global dollar strength is controlled—further evidence that stablecoin metrics partially proxy dollar conditions. Fourth, when the dependent variable is the absolute FX change, only DAI circulation is (negatively) significant, suggesting stablecoin growth is not associated with higher FX volatility. As a final cross-check, re-estimating the by-coin baseline with the `statspai` package reproduces the `linearmodels` coefficients to three decimals.

第一，周度频率下所有币种系数均不显著：微小的日度联动无法经加总保留，说明稳定币—汇率的任何关联都是短周期现象。第二，剔除俄罗斯及 Terra/SVB 危机月份后，USDC 流通量系数仍显著为负（ -0.0051 ），表明基准结果并非由这些事件驱动。第三，以 DXY 替代广义美元指数后，USDT 系数转为边际显著的负值（ -0.0051 ），凸显币种系数对美元整体走势控制方式的敏感性——这进一步说明稳定币指标部分地代理了美元环境。第四，以汇率变化绝对值为被解释变量时，仅 DAI 流通量（负向）显著，说明稳定币增长与汇率波动上升无关。作为最后的交叉核验，用 `statspai` 包重新估计的分币种基准回归与 `linearmodels` 的系数在三位小数上一致。

8. Conclusion / 结论

This paper provides a currency-by-currency analysis of how USD-pegged stablecoins relate to exchange rates, using a 27-economy daily panel for 2020–2025 with four on-chain metrics per coin, country-characteristic interactions, and a crypto-shock IV design.

本文利用 2020–2025 年 27 个经济体的日度面板、每种币的四项链上指标、国家特征交互项与加密冲击工具变量设计，对美元稳定币与汇率的关系进行了逐币种分析。

The findings temper the view that stablecoins are a first-order driver of exchange rates in normal times. Average effects of circulation, issuance, volume, and wallet growth on daily FX changes are small and mostly insignificant; the only robust baseline relationship is a modest *appreciation* associated with USDC circulation growth. Heterogeneity runs opposite to the usual narrative: USDT matters in developed and high-crypto-adoption economies rather than in emerging markets, and moderation by country characteristics is weak. The IV estimates—with strong first stages—detect no causal average effect at the daily horizon.

研究发现对”稳定币在正常时期是汇率一阶驱动因素”的观点构成了降温。流通量、发行量、交易量与钱包增长对日度汇率变化的平均效应很小且大多不显著；唯一稳健的基准关系是 USDC 流通量增长伴随小幅升值。异质性方向与惯常叙事相反：USDT 的影响体现在发达经济体和高加密采用经济体，而非新兴市场；国家特征的调节作用很弱。第一阶段很强的工具变量估计在日度频率上未检测到因果平均效应。

These results do not rule out crisis-time spillovers or effects at longer horizons; rather, they indicate that such effects cannot be extrapolated to daily normal times. For policy, the implication is that monitoring stablecoin activity is most warranted where crypto adoption and market infrastructure are deepest [13], and that claims of pervasive stablecoin-driven depreciation pressure in emerging markets require evidence at frequencies or episodes beyond normal daily variation. Future work could exploit higher-frequency intraday data, crisis episodes such as exchange market pressure events, or issuer-level geography to trace the channels more precisely [5].

这些结果并不排除危机时期或更长 horizon 上的溢出效应；它们表明此类效应不能外推至正常时期的日度变化。政策含义在于：最应当在加密采用度和市场基础设施最深的经济体监测稳定币活动 [13]；而”稳定币在新兴市场造成普遍贬值压力”的主张，需要超出正常日度波动的频率或事件层面的证据支持。未来研究可利用更高频的盘中数据、汇率市场压力事件等危机情景或发行人层面的地理分布，更精确地追踪传导渠道 [5]。

Appendix A. Data Construction Details / 数据构造细节

Country sample / 国家样本. The 27 economies are: Argentina, Australia, Brazil, Canada, Chile, China, Colombia, the euro area, India, Indonesia, Japan, Korea, Malaysia, Mexico, Nigeria, Norway, the Philippines, Poland, Russia, Saudi Arabia, Singapore, South Africa, Sweden, Switzerland, Thailand, Türkiye, and the United Kingdom. Eleven are classified as developed (EUR, GBP, JPY, CAD, AUD, CHF, SEK, NOK, SGD, KRW, PLN) and sixteen as emerging markets.

27 个经济体为：阿根廷、澳大利亚、巴西、加拿大、智利、中国、哥伦比亚、欧元区、印度、印度尼西亚、日本、韩国、马来西亚、墨西哥、尼日利亚、挪威、菲律宾、波兰、俄罗斯、沙特阿拉伯、新加坡、南非、瑞典、瑞士、泰国、土耳其和英国。其中 11 个归为发达经济体 (EUR、GBP、JPY、CAD、AUD、CHF、SEK、NOK、SGD、KRW、PLN)，16 个归为新兴市场。

Exchange-rate splicing / 汇率拼接. RUB is spliced from ECB reference rates (through February 2022) and Yahoo Finance (from March 2022), with one erroneous Yahoo tick (16 March 2023) removed. AUD, IDR, PHP, PLN, THB, and TRY are converted from EUR crosses using FRED’s USD-per-EUR rate. ARS, CLP, COP, NGN, and SAR are sourced directly from Yahoo Finance. All series are expressed as local currency per USD.

RUB 由欧洲央行参考汇率（至 2022 年 2 月）与雅虎财经（自 2022 年 3 月）拼接而成，并剔除了雅虎的一个错误报价点（2023 年 3 月 16 日）。AUD、IDR、PHP、PLN、THB、TRY 通过欧元交叉汇率经 FRED 美元兑欧元报价换算。ARS、CLP、COP、NGN、SAR 直接取自雅虎财经。所有序列均以本币/美元表示。

Stablecoin metrics / 稳定币指标. Circulation is Glassnode circulating supply; issuance is the daily first difference of supply; volume and wallets are Glassnode aggregated transfer volume and active addresses. Issuance growth uses the signed log transform $\Delta[\text{sign}(I_t) \ln(1 + |I_t|)]$ to handle negative net issuance.

流通量为 Glassnode 流通供给量；发行量为供给量的日度一阶差分；交易量与钱包数为 Glassnode 汇总转账交易量与活跃地址数。发行量增速采用带符号对数变换 $\Delta[\text{sign}(I_t) \ln(1 + |I_t|)]$ 以处理净发行为负的情形。

Country characteristics / 国家特征. Annual characteristics are lagged by one year. The euro area is treated as a single economy: its Chinn–Ito value is the mean of Germany, France, Italy, Spain, and the Netherlands; its government debt and fiscal balance are Eurostat EA20 aggregates; its current account is the IMF DataMapper euro-area aggregate; and its IRR classification is set to freely floating. Short gaps in WDI series are closed by within-country carry-forward, and the Chinn–Ito index and latest IRR classification are held constant beyond their last available years.

国家特征滞后一年。欧元区作为单一经济体处理：其 Chinn–Ito 取值为德国、法国、意大利、西班牙和荷兰的均值；政府债务与财政余额为 Eurostat EA20 汇总值；经常账户为 IMF DataMapper 欧元区汇总值；IRR 分类设为自由浮动。WDI 序列的短期缺口按国别前值结转填补，Chinn–Ito 指数与最新 IRR 分类在其最后可得年份之后保持不变。

Reproducibility / 可复现性. All results are produced by the notebook `stablecoin_fx_panel.ipynb` (conda environment `ci`), which reads `../Countries Dataset/panel_daily.csv` (built by `work/build_panel.py`) and writes the figures and tables used in this paper to `figures/` and `tables/`. Estimation uses `linearmodels` (PanelOLS, IV2SLS) with a `statspai` cross-check.

全部结果由 notebook `stablecoin_fx_panel.ipynb` (conda 环境 `ci`) 生成，其读取 `../Countries Dataset/panel_daily.csv` (由 `work/build_panel.py` 构建)，并将本文所用图表写入 `figures/` 与 `tables/`。估计使用 `linearmodels` (PanelOLS、IV2SLS)，并以 `statspai` 交叉核验。

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Table 1: Summary statistics / 描述性统计. Daily variables in percent, 2020–2025; Panel F reports the 2023 cross-section of country characteristics.

	Mean	Std. Dev.	Min	Max
<i>Panel A: FX change (%) / 汇率变化</i>				
Full sample / 全样本	0.012	0.687	−5.781	7.381
Emerging markets / 新兴市场	0.021	0.767	−5.781	7.381
Developed economies / 发达经济体	−0.001	0.556	−1.970	2.226
<i>Panel B: USDT / USDT 指标</i>				
Circulation / 流通量	0.196	1.193	−3.004	38.934
Issuance / 发行量	0.555	23.405	−124.345	125.124
Trading volume / 交易量	10.596	33.070	−133.633	224.190
Active wallets / 活跃钱包	2.923	10.056	−45.183	63.932
<i>Panel C: USDC / USDC 指标</i>				
Circulation / 流通量	0.178	1.115	−8.563	10.011
Issuance / 发行量	0.421	24.905	−183.665	143.060
Trading volume / 交易量	7.474	36.665	−245.715	226.771
Active wallets / 活跃钱包	4.057	15.273	−80.839	131.915
<i>Panel D: DAI / DAI 指标</i>				
Circulation / 流通量	0.188	2.960	−56.441	59.005
Issuance / 发行量	−0.119	8.495	−115.201	97.522
Trading volume / 交易量	7.999	72.687	−393.834	408.997
Active wallets / 活跃钱包	2.901	15.144	−70.304	110.414
<i>Panel E: Controls / 控制变量</i>				
VIX (% change / 变动)	−0.120	7.877	−44.245	55.411
USD index (% change / 美元指数变动)	−0.001	0.330	−1.910	1.875
<i>Panel F: Country characteristics, 2023 / 国家特征 (2023 年)</i>				
Capital openness (Chinn–Ito) / 资本开放度	0.606	0.352	0.162	1.000
Fixed regime (IRR) / 固定汇率制度	0.222	0.424	0.000	1.000
Reserves / months of imports / 储备覆盖进口月数	6.463	4.363	1.254	16.569
Gov. debt / GDP (%) / 政府债务占 GDP	70.446	47.234	15.190	220.302
Fiscal balance / GDP (%) / 财政余额占 GDP	−2.521	4.513	−8.374	15.858
Current account / GDP (%) / 经常账户占 GDP	1.537	5.113	−3.634	17.171
Income level (log GDP p.c.) / 收入水平 (人均 GDP 对数)	9.808	1.111	7.668	11.556
Crypto adoption (Chainalysis) / 加密采用指数	0.302	0.320	0.000	1.000
Observations / 观测数	37,370			
Countries / 国家数	27			
Time period / 样本期	2020–2025			

Table 2: Baseline results / 基准结果: stablecoins and daily exchange-rate changes (%). Country fixed effects and global controls (VIX and broad USD index changes) included throughout.

Dependent variable / 被解释变量:	FX change / 汇率变化 (%)		
	(1) Aggregate / 汇总	(2) By coin / 分币种	(3) All metrics / 全指标
Aggregate circulation growth / 总流通量增速	0.0035 (0.0035)		
USDT: circulation / 流通量		0.0037 (0.0028)	0.0047 (0.0035)
USDT: issuance / 发行量			-0.0001 (0.0002)
USDT: trading volume / 交易量			-0.0002 (0.0002)
USDT: active wallets / 活跃钱包			0.0002 (0.0004)
USDC: circulation / 流通量		-0.0035 (0.0023)	-0.0063** (0.0029)
USDC: issuance / 发行量			0.0002 (0.0002)
USDC: trading volume / 交易量			0.0002 (0.0001)
USDC: active wallets / 活跃钱包			0.0000 (0.0002)
DAI: circulation / 流通量		0.0004 (0.0010)	-0.0005 (0.0012)
DAI: issuance / 发行量			0.0006* (0.0003)
DAI: trading volume / 交易量			-0.0002*** (0.0001)
DAI: active wallets / 活跃钱包			-0.0000 (0.0004)
VIX change / VIX 变动	0.0010 (0.0012)	0.0010 (0.0012)	0.0012 (0.0013)
Broad USD index / 广义美元指数	0.9165*** (0.0948)	0.9170*** (0.0950)	0.9203*** (0.0949)
Country fixed effects / 国家固定效应	Yes	Yes	Yes
Global controls / 全球控制	Yes	Yes	Yes
Observations / 观测数	37,370	37,370	37,370
Countries / 国家数	27	27	27
R^2	0.197	0.197	0.198

Notes / 注: Standard errors clustered by country in parentheses. *, **, *** denote significance at 10%, 5%, 1%. 括号内为按国家聚类的标准误; *, **, *** 分别表示 10%、5%、1% 显著性水平。

Table 3: Interaction regressions / 交互项回归: USDT circulation growth interacted with standardized country characteristics (2023 values).

Characteristic / 特征	USDT main / 主效应	USDT $\times$ Char. / 交互项
Capital openness (Chinn-Ito) / 资本开放度	0.0037 (0.0028)	-0.0024 (0.0026)
Fixed regime (IRR) / 固定汇率制度	0.0037 (0.0029)	0.0030 (0.0029)
Reserves / months of imports / 储备覆盖	0.0037 (0.0028)	0.0037* (0.0019)
Gov. debt / GDP / 政府债务	0.0037 (0.0029)	0.0004 (0.0026)
Fiscal balance / GDP / 财政余额	0.0037 (0.0028)	-0.0024* (0.0013)
Current account / GDP / 经常账户	0.0037 (0.0028)	0.0007 (0.0025)
Income level (log GDP p.c.) / 收入水平	0.0037 (0.0028)	-0.0021 (0.0034)
Crypto adoption (Chainalysis) / 加密采用度	0.0037 (0.0027)	0.0041 (0.0032)
Observations / 观测数	37,370	
Countries / 国家数	27	

Notes / 注: Each row is a separate regression including USDT, USDC and DAI circulation growth and global controls, with country fixed effects; standard errors clustered by country. 每行为单独回归, 均包含三种币流通量增速与全球控制变量及国家固定效应; 标准误按国家聚类。

Table 4: Crypto-shock IV estimates / 加密冲击工具变量估计: 2SLS with Terra/LUNA, SVB de-peg, and lagged Ethereum gas-price instruments.

Dependent variable / 被解释变量:	FX change / 汇率变化 (%)		
	(1) USDT	(2) USDC	(3) DAI
Circulation growth (instrumented) / 流通量增速 (工具化)	-0.0190 (0.0593)	0.0241 (0.0259)	0.0211 (0.0183)
First-stage F (excluded instruments) / 第一阶段 F	74.02	248.25	75.26
Country fixed effects (within) / 国家固定效应	Yes	Yes	Yes
Global controls / 全球控制	Yes	Yes	Yes
Observations / 观测数	37,343	37,343	37,343
Countries / 国家数	27	27	27

Notes / 注: Instruments: Terra/LUNA window (5–20 May 2022), SVB de-peg window (9–17 March 2023), lagged mean Ethereum gas-price change. Standard errors clustered by country. 工具变量: Terra/LUNA 窗口 (2022 年 5 月 5–20 日)、SVB 脱锚窗口 (2023 年 3 月 9–17 日)、滞后以太坊 Gas 价格变动。标准误按国家聚类。

Table 5: Subsample regressions by country group / 分组子样本回归: circulation-growth coefficients.

Group / 分组	USDT	USDC	DAI	N	Countries / 国家数	R^2
Emerging markets / 新兴市场	0.0024 (0.0042)	-0.0016 (0.0034)	-0.0004 (0.0017)	21,736	16	0.104
Developed economies / 发达经济体	0.0057* (0.0033)	-0.0060** (0.0027)	0.0013* (0.0007)	15,634	11	0.483
Closed capital account / 资本账户封闭	0.0056 (0.0045)	-0.0028 (0.0038)	-0.0012 (0.0018)	17,822	13	0.097
Open capital account / 资本账户开放	0.0020 (0.0035)	-0.0040 (0.0027)	0.0018* (0.0010)	19,548	14	0.374
Fixed regime (IRR) / 固定汇率制	0.0061 (0.0052)	-0.0053** (0.0025)	-0.0047** (0.0018)	8,172	6	0.075
Floating regime (IRR) / 浮动汇率制	0.0030 (0.0033)	-0.0030 (0.0029)	0.0017 (0.0011)	29,198	21	0.235
Low reserves / 低储备	0.0044 (0.0040)	-0.0067** (0.0028)	-0.0000 (0.0013)	23,630	17	0.255
High reserves / 高储备	0.0026 (0.0037)	0.0019 (0.0035)	0.0011 (0.0017)	13,740	10	0.119
High government debt / 高政府债务	0.0052 (0.0049)	-0.0039 (0.0041)	0.0010 (0.0008)	14,044	10	0.291
Low government debt / 低政府债务	0.0029 (0.0034)	-0.0033 (0.0028)	-0.0000 (0.0016)	23,326	17	0.169
High crypto adoption / 高加密采用	0.0098** (0.0042)	-0.0025 (0.0043)	-0.0013 (0.0021)	15,330	11	0.133
Low crypto adoption / 低加密采用	-0.0005 (0.0035)	-0.0042 (0.0026)	0.0016* (0.0009)	22,040	16	0.277

Notes / 注: Each row is a separate by-coin regression with country fixed effects and global controls; standard errors clustered by country. 每行为单独的分币种回归, 含国家固定效应与全球控制; 标准误按国家聚类。

Table 6: Robustness checks / 稳健性检验: circulation-growth coefficients under alternative samples, controls, and outcomes.

Specification / 设定	USDT	USDC	DAI	N
Weekly frequency / 周度频率	0.0037 (0.0055)	-0.0029 (0.0033)	-0.0005 (0.0015)	8,478
Excl. RUS & crisis months / 剔除俄罗斯与危机月份	0.0041 (0.0030)	-0.0051** (0.0024)	0.0000 (0.0010)	34,975
DXY instead of broad USD index / 以 DXY 替代广义美元指数	-0.0051* (0.0028)	-0.0043* (0.0024)	-0.0014 (0.0009)	37,370
Dep. var = $ \Delta s $ / 被解释变量为汇率波动	-0.0015 (0.0034)	0.0020 (0.0028)	-0.0039*** (0.0010)	37,370

Notes / 注: Country fixed effects and global controls included; standard errors clustered by country. Crisis months are May 2022 and March 2023. 均含国家固定效应与全球控制; 标准误按国家聚类; 危机月份为 2022 年 5 月与 2023 年 3 月。